Java Full Stack Development

1. Spring Data

Module Topics

- Spring Data Design
- Repository Pattern
- Entity Mappings
- Spring JPA
- Transactions
- Spring Data LDAP
- Spring Data Redis
- Spring Data Rest

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Spring Data Design

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Spring Data

- The traditional data access problem
  - Every interaction with a data store requires boilerplate
    - Obtaining a connection
    - Preparing a statement
    - Executing the statement
    - Mapping results
    - Handling exceptions
    - Releasing resources
  - Boilerplate
    - Is repetitive, error-prone
    - Clutters the application with infrastructure concerns that have nothing to do with business logic

```java
// Raw JDBC -- every operation requires this scaffolding
public Employee findById(Long id) {
    Connection conn = null;
    PreparedStatement stmt = null;
    ResultSet rs = null;
    try {
        conn = dataSource.getConnection();
        stmt = conn.prepareStatement(
            "SELECT employee_id, full_name, salary FROM employees WHERE employee_id = ?"
        );
        stmt.setLong(1, id);
        rs = stmt.executeQuery();
        if (rs.next()) {
            return new Employee(
                rs.getLong("employee_id"),
                rs.getString("full_name"),
                rs.getBigDecimal("salary")
            );
        }
        return null;
    } catch (SQLException e) {
        throw new RuntimeException("Database error", e);
    } finally {
        // Must close in reverse order -- if any close throws, others are skipped
        if (rs   != null) try { rs.close();   } catch (SQLException ignored) {}
        if (stmt != null) try { stmt.close(); } catch (SQLException ignored) {}
        if (conn != null) try { conn.close(); } catch (SQLException ignored) {}
    }
}
```

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Spring Data

- The traditional data access problem
  - Different data stores each have their own APIs, connection models, and error handling conventions
    - Relational databases
    - Key-value stores
    - Document stores
    - Directory services
  - This means the application code is tightly coupled to the specific data store technology
  - And also tightly coupled to the underlying data model
  - The resulting system is very brittle

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        throw new RuntimeException("Database error", e);
    } finally {
        // Must close in reverse order -- if any close throws, others are skipped
        if (rs   != null) try { rs.close();   } catch (SQLException ignored) {}
        if (stmt != null) try { stmt.close(); } catch (SQLException ignored) {}
        if (conn != null) try { conn.close(); } catch (SQLException ignored) {}
    }
}
```

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Spring Data

- Design goal of Spring Data
  - Decouple the boilerplate scaffolding from the application code
    - Declarative statement of the interface to the repository
    - Let the framework generate the boilerplate code
  - Data access operations follow predictable patterns
    - Find by ID, find all, save, delete, find by criteria
    - These patterns are the same regardless of whether the store is Oracle, Redis, or LDAP
    - Spring Data codifies these patterns into reusable abstractions so developers declare what they need rather than implementing it each time

```java
// The entire above method is replaced by this single interface declaration
public interface EmployeeRepository extends CrudRepository<Employee, Long> {
    // findById(Long id) is already provided by CrudRepository
}
```

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Design Goals

- 1: Reduce boilerplate to the minimum necessary expression of intent
  - The developer declares what data operations the application needs
  - The framework provides implementations of those operations
  - The developer does not implement what the framework already knows how to do
- 2: Provide a consistent programming model across different data stores
  - The same repository abstractions, the same annotation model, and the same query derivation mechanism work whether the underlying store is a relational database, a key-value store, a document store, or a directory service
  - Switching from one store to another changes configuration and dependencies but not the application's data access logic

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Design Goals

- 3: Support store-specific features without hiding them
  - Spring Data does not force a lowest-common-denominator abstraction that loses the power of specific stores
    - Oracle-specific SQL can be expressed with @Query; Redis pipeline operations are available; LDAP filter syntax is supported
    - The abstraction reduces boilerplate but never prevents access to the full capability of the underlying store
- 4: Integrate naturally with the Spring ecosystem
  - Repositories are Spring beans
    - They participate in dependency injection, transaction management, and the application context lifecycle
  - Spring Data works with Spring's transaction manager so that data access operations participate in the same transactions as service layer logic
  - Configuration follows Spring Boot's auto-configuration model
    - Adding a dependency and providing connection properties is usually sufficient to get a working data access layer

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Repository Pattern

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Repository Pattern

- The Repository pattern
  - Described by Eric Evans in Domain-Driven Design (2003)
- A repository
  - An object that mediates between the domain layer and the data mapping layer
  - Presents the application with a collection-like interface for accessing domain objects
  - The application asks the repository for objects as if they were in a collection in memory
  - The repository handles the mechanics of retrieval from the actual store

```
Repository<T, ID>                    -- marker interface, no methods
  └── CrudRepository<T, ID>          -- save, findById, findAll, delete, count, exists
        └── PagingAndSortingRepository<T, ID>  -- findAll with Sort and Pageable
              └── JpaRepository<T, ID>         -- JPA-specific: flush, saveAndFlush, batch operations
```

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Repository Pattern

- Declaring a repository
  - T is the domain type (the entity class)
  - ID is the type of the entity's primary key
  - Each level adds capabilities to the one above it
  - The application extends the interface appropriate for its needs
    - It does not implement anything

```
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```

```java
// Extend the appropriate interface for the required capability level
public interface EmployeeRepository extends JpaRepository<Employee, Long> {

    // Spring Data derives a query from the method name automatically
    List<Employee> findByDepartmentIdAndStatus(Long departmentId, String status);

    // Spring Data derives an ORDER BY from the method name
    List<Employee> findByDepartmentIdOrderBySalaryDesc(Long departmentId);

    // For more complex queries, provide the JPQL explicitly
    @Query("SELECT e FROM Employee e WHERE e.salary > :threshold AND e.status = 'ACTIVE'")
    List<Employee> findHighEarners(@Param("threshold") BigDecimal threshold);
}
```

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Repository Pattern

- Spring Data
  - Provides the implementation
  - At application startup, Spring Data scans for interfaces that extend Repository
  - For each one, it generates a proxy implementation using Java reflection and the JDK proxy mechanism
  - The proxy implements every method: those inherited from the parent interface and those declared in the application interface
  - The proxy is registered as a Spring bean and is available for injection throughout the application

```
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Repository Pattern

- What the JDK proxy mechanism is
  - Facility in the Java standard library
    - Creates a concrete object implementing one or more interfaces at runtime without any hand-written class
    - The created object is the proxy.
  - Two things are required to create a JDK proxy
    - The interface or interfaces it should implement
    - An InvocationHandler, a single method that receives every method call made on the proxy and decides what to do with it
  - A proxy is an object that stands in for another object and intercepts calls made to it

```java
// This is conceptually what Spring Data does internally
// (simplified -- the real implementation is considerably more sophisticated)

InvocationHandler handler = (proxy, method, args) -> {
    // Every call to any method on the repository arrives here
    // method.getName() tells us which method was called
    // args contains the arguments

    if (method.getName().equals("findById")) {
        // Generate and execute: SELECT * FROM employees WHERE employee_id = ?
        // Map the ResultSet to an Employee object
        // Return it
    }

    if (method.getName().startsWith("findBy")) {
        // Parse the rest of the method name to derive the WHERE clause
        // Generate and execute the appropriate SQL
        // Map the results
    }

    if (method.isAnnotationPresent(Query.class)) {
        // Read the JPQL or SQL from the @Query annotation
        // Bind the parameters
        // Execute and map the results
    }
    // ... and so on for every supported pattern
};

Object repositoryProxy = Proxy.newProxyInstance(
    classLoader,
    new Class[]{ EmployeeRepository.class },
    handler
);
```

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Repository Pattern

- In Spring Data
  - The developer declares a repository interface but writes no implementation class
  - At application startup, Spring Data needs something concrete that can be injected wherever that interface is used
  - The proxy is that concrete object

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Repository Pattern

- Java reflection
  - Capabilities built into the Java runtime that allows code to inspect and interact with the structure of other code at runtime rather than at compile time
  - Through reflection, a running program can
    - Discover what methods an interface declares, without knowing the interface at compile time
    - Read annotations on those methods and their parameters
    - Create new objects and invoke methods dynamically, by name, without a direct call in the source code

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Repository Pattern

- Java reflection
  - Spring Data uses reflection to inspect the repository interface the developer declared
  - Reads the interface name, the methods declared on it, the parent interfaces it extends, and all the annotations on those methods
  - This inspection happens at startup and gives Spring Data everything it needs to understand what the repository is supposed to do.

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Repository Pattern

- How Spring Data works
  - At application startup, Spring Data scans for interfaces that extend Repository
  - For each one, it generates a proxy implementation using Java reflection and the JDK proxy mechanism
  - The proxy implements every method
    - Those inherited from the parent interface and those declared in the application interface
  - The proxy is registered as a Spring bean and is available for injection throughout the application

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Repository Pattern

- CrudRepository
  - Sufficient for most operations
  - PagingAndSortingRepository
    - Adds the ability to retrieve data in pages and in a specified order without writing any pagination logic
  - JpaRepository
    - Adds JPA-specific operations that are not in the generic interfaces
    - An application that only needs basic CRUD should extend CrudRepository there is no reason to pull in JPA-specific capabilities if they are not needed

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    // For more complex queries, provide the JPQL explicitly
    @Query("SELECT e FROM Employee e WHERE e.salary > :threshold AND e.status = 'ACTIVE'")
    List<Employee> findHighEarners(@Param("threshold") BigDecimal threshold);
}
```

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Repository Pattern

- Spring Data
  - Parses the method name and constructs a query from it
    - findByDepartmentIdAndStatus becomes a query with a WHERE clause on department_id and status
    - findByDepartmentIdOrderBySalaryDesc adds an ORDER BY salary DESC
  - Works well for simple queries and eliminates the need to write boilerplate query strings for common lookup patterns
  - For complex queries, the @Query annotation provides an escape hatch
    - Developer writes the query explicitly and Spring Data handles binding parameters and mapping results

```
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```

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Query Derivation

- Turning method names into queries
  - Spring Data parses repository method names according to a grammar
  - The method name is split into
    - A subject or what to do
    - And a predicate or the conditions
  - Spring Data constructs the appropriate query for the underlying store from this parsed structure
  - This is called method name derivation

Method name grammar

```
find...By...     -- SELECT query
count...By...    -- COUNT query
exists...By...   -- EXISTS check returning boolean
delete...By...   -- DELETE statement

Conditions:
And              -- AND between two conditions
Or               -- OR between two conditions
Between          -- BETWEEN lower AND upper
LessThan         -- < value
GreaterThan      -- > value
Like             -- LIKE pattern
NotLike          -- NOT LIKE pattern
In               -- IN (collection)
NotIn            -- NOT IN (collection)
IsNull           -- IS NULL
IsNotNull        -- IS NOT NULL
OrderBy          -- ORDER BY column Asc/Desc
```

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Query Derivation

- Method name derivation
  - Powerful feature for simple lookups
    - Not a replacement for SQL or JPQL for anything beyond straightforward single-table queries
    - Natural fit is simple find-by-field operations
    - More complex queries should use explicit @Query
  - Rules of thumb
    - If the method name requires reading twice to understand what query it represents, replace it with @Query with a clear name and an explicit query string
    - Method names become unreadable beyond two or three conditions
    - Any query involving joins, subqueries, aggregations, or complex expressions requires @Query

```java
// WHERE status = ? AND department_id = ?
List<Employee> findByStatusAndDepartmentId(String status, Long departmentId);

// WHERE salary BETWEEN ? AND ?
List<Employee> findBySalaryBetween(BigDecimal min, BigDecimal max);

// WHERE full_name LIKE ?  (caller must supply the % characters)
List<Employee> findByFullNameLike(String pattern);

// WHERE department_id IN (?, ?, ...)
List<Employee> findByDepartmentIdIn(Collection<Long> departmentIds);

// WHERE status IS NOT NULL ORDER BY hire_date DESC
List<Employee> findByStatusIsNotNullOrderByHireDateDesc();

// Returns true if any matching row exists
boolean existsByEmailAddress(String email);

// Returns the count of matching rows
long countByStatus(String status);
```

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Query Derivation

- Method name derivation
  - The Like condition
    - Spring Data does not add the % wildcard characters automatically
    - The caller must supply the pattern with % characters included
  - The Contains variant that wraps the value in % automatically
    - StartingWith and EndingWith used for prefix and suffix matches
    - These are more readable than Like for the common cases
    - FindByFullNameContaining has LIKE functionality

```java
// WHERE status = ? AND department_id = ?
List<Employee> findByStatusAndDepartmentId(String status, Long departmentId);

// WHERE salary BETWEEN ? AND ?
List<Employee> findBySalaryBetween(BigDecimal min, BigDecimal max);

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Entity Mappings

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Entity Mapping

- In Spring Data
  - A Java class is the entity
    - Represents a table row, a document, a cache entry, or a directory object
  - Annotations on the class and its fields tell Spring Data how to map between the Java object and the underlying store
  - The choice of annotations depends on the Spring Data module being used

```java
@Entity                                  // marks this class as a JPA-managed entity
@Table(name = "employees")               // maps to the employees table
public class Employee {

    @Id                                  // marks this field as the primary key
    @GeneratedValue(strategy = GenerationType.IDENTITY)  // uses the DB identity column
    @Column(name = "employee_id")
    private Long employeeId;

    @Column(name = "full_name", nullable = false, length = 200)
    private String fullName;

    @Column(name = "salary", precision = 15, scale = 2)
    private BigDecimal salary;

    @Column(name = "hire_date")
    private LocalDate hireDate;

    @ManyToOne(fetch = FetchType.LAZY)   // lazy: do not load department unless accessed
    @JoinColumn(name = "department_id")
    private Department department;

    // getters and setters omitted
}
```

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JDBC Mapping

- JPA
  - Uses @Entity and supports bi-directional relationships, lazy loading, and a managed entity lifecycle
- Spring Data JDBC
  - Uses @Table only, has no lazy loading, and models relationships through aggregate roots rather than navigable object references
  - These are not just syntax differences
    - They reflect fundamentally different approaches to the object-relational mapping problem

```java
@Table("employees")                    // maps to the employees table
public class Employee {

    @Id
    private Long employeeId;           // mapped to employee_id by naming convention

    private String fullName;           // mapped to full_name by naming convention
    private BigDecimal salary;
    private LocalDate hireDate;

    // No relationship annotations -- Spring Data JDBC uses aggregate roots
    // instead of bi-directional object graph navigation
}
```

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JDBC Mapping

- Spring Data JDBC
  - Lightweight relational data access module that maps Java objects to relational tables without a full ORM layer
  - Does not use JPA or Hibernate
    - It has its own mapping layer built on Spring's JdbcTemplate
  - It follows the Aggregate Root pattern from Domain-Driven Design
    - Each repository manages one aggregate
    - Objects within the aggregate are loaded and saved together as a unit

```java
// JPA approach -- navigable object reference
@ManyToOne(fetch = FetchType.LAZY)
@JoinColumn(name = "department_id")
private Department department;   // holds a Department object (or a proxy for one)

// Usage: traverses the object graph, may trigger a database query silently
String deptName = employee.getDepartment().getDepartmentName();


// Spring Data JDBC approach -- aggregate boundary, ID reference only
@Column("department_id")
private Long departmentId;       // holds an ID, not an object

// Usage: explicit repository call -- the database access is visible
Department dept = departmentRepository.findById(employee.getDepartmentId())
                      .orElseThrow();
String deptName = dept.getDepartmentName();
```

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JDBC Mapping

- The aggregate root model
  - An aggregate is a cluster of objects that belong together and are treated as a unit for data access purposes
  - The aggregate root is the top-level object in that cluster
    - The repository operates on the root, not on individual parts
  - When the root is saved, everything inside it is saved
  - When it is loaded, everything inside it is loaded
  - There is no lazy loading
    - If the data is part of the aggregate, it is always loaded with it

```java
// Employee is the aggregate root
// Address is part of the Employee aggregate
@Table("employees")
public class Employee {

    @Id
    private Long employeeId;
    private String fullName;
    private BigDecimal salary;

    @Embedded(onEmpty = Embedded.OnEmpty.USE_NULL)
    private Address address;   // stored in the employees table, not a separate entity
}

// Repository operates on the aggregate root
public interface EmployeeRepository extends CrudRepository<Employee, Long> {
    List<Employee> findByAddressCity(String city);
}
```

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JDBC Mapping

- When Spring Data JDBC is the right choice
  - The domain model is relatively simple and maps cleanly to a relational schema
  - The application needs predictable, explicit SQL behaviour without ORM magic
  - Performance is important and the overhead of JPA's managed entity lifecycle is not justified
  - The team values simplicity and explicit control over convenience and automation

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// Address is part of the Employee aggregate
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// Repository operates on the aggregate root
public interface EmployeeRepository extends CrudRepository<Employee, Long> {
    List<Employee> findByAddressCity(String city);
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Spring JPA

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Spring Data JPA

- Spring Data JPA
  - The most feature-rich relational data access module in Spring Data
  - Provides the Spring Data repository abstraction on top of the Java Persistence API (JPA)
    - Using Hibernate as the default JPA provider
  - Manages the full lifecycle of entity objects: the application works with Java objects and JPA translates those operations to SQL automatically

```java
public interface EmployeeRepository extends JpaRepository<Employee, Long> {

    // JPQL query -- operates on entity class names and field names, not table/column names
    @Query("SELECT e FROM Employee e JOIN FETCH e.department " +
            "WHERE e.department.id = :deptId AND e.status = :status")
    List<Employee> findWithDepartmentByDeptIdAndStatus(
            @Param("deptId") Long deptId,
            @Param("status") String status
    );

    // Native SQL query for Oracle-specific syntax
    @Query(value = "SELECT * FROM employees WHERE ROWNUM <= :limit " +
            "ORDER BY salary DESC",
            nativeQuery = true)
    List<Employee> findTopEarners(@Param("limit") int limit);

    // Derived query with pagination -- Spring Data handles the OFFSET/FETCH automatically
    Page<Employee> findByStatus(String status, Pageable pageable);
}
```

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Spring Data JPA

- The JPA entity lifecycle
  - Transient
    - The object exists in Java but is not associated with any database row
  - Persistent (managed)
    - The object is associated with a database row and tracked by the JPA session (EntityManager)
    - Any change to the object is automatically detected and written to the database at the end of the transaction
  - Detached
    - The object was previously managed but the session has closed -- changes are not tracked
  - Removed
    - The object is marked for deletion and will be deleted at the end of the transaction

```java
public interface EmployeeRepository extends JpaRepository<Employee, Long> {

    // JPQL query -- operates on entity class names and field names, not table/column names
    @Query("SELECT e FROM Employee e JOIN FETCH e.department " +
            "WHERE e.department.id = :deptId AND e.status = :status")
    List<Employee> findWithDepartmentByDeptIdAndStatus(
            @Param("deptId") Long deptId,
            @Param("status") String status
    );

    // Native SQL query for Oracle-specific syntax
    @Query(value = "SELECT * FROM employees WHERE ROWNUM <= :limit " +
            "ORDER BY salary DESC",
            nativeQuery = true)
    List<Employee> findTopEarners(@Param("limit") int limit);

    // Derived query with pagination -- Spring Data handles the OFFSET/FETCH automatically
    Page<Employee> findByStatus(String status, Pageable pageable);
}
```

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Spring Data JPA

- What JPA provides over Spring Data JDBC
  - Lazy loading
    - Related objects are not loaded from the database until they are accessed
  - Dirty checking
    - JPA detects changes to managed entities and generates UPDATE statements automatically without explicit save calls
  - First-level cache
    - Within a session, JPA returns the same object instance for the same primary key, reducing database reads
  - JPQL
    - an object-oriented query language that operates on entity classes rather than tables
  - Cascading
    - Operations on a parent entity can automatically cascade to child entities

```java
public interface EmployeeRepository extends JpaRepository<Employee, Long> {

    // JPQL query -- operates on entity class names and field names, not table/column names
    @Query("SELECT e FROM Employee e JOIN FETCH e.department " +
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            "ORDER BY salary DESC",
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    // Derived query with pagination -- Spring Data handles the OFFSET/FETCH automatically
    Page<Employee> findByStatus(String status, Pageable pageable);
}
```

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Comparison

- Fundamental differences
  - Spring Data JDBC follows the principle that
    - SQL is the right language for relational data
    - The framework's job is to reduce the boilerplate of executing it
  - Spring Data JPA follows the principle that
    - The application should work with objects
    - The framework should handle the translation to SQL automatically

Dimension	Spring Data JDBC	Spring Data JPA
Underlying technology	Spring JdbcTemplate -- no ORM	JPA (Hibernate) -- full ORM
Entity lifecycle	None -- objects are plain Java objects	Managed -- JPA tracks changes to persistent entities
Lazy loading	Not supported -- load the whole aggregate	Supported -- related objects loaded on access
Dirty checking	Not supported -- must call save() explicitly	Supported -- changes detected automatically
Query language	SQL (via @Query) or derived methods	JPQL or SQL (via @Query with nativeQuery)
Performance predictability	High -- SQL is explicit	Lower -- lazy loading and dirty checking generate implicit SQL
Learning curve	Low -- minimal new concepts	Higher -- entity lifecycle, JPQL, caching, cascading
Best fit	Simpler domains, performance-sensitive code, explicit SQL preference	Complex domain models, rich object graph navigation, rapid development

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Comparison

- Practical rule
  - Start with Spring Data JDBC
    - It is simpler, more predictable, and sufficient for most use cases
    - Move to Spring Data JPA when the domain model is complex enough that managing object relationships explicitly becomes burdensome
  - Use native queries via @Query(nativeQuery = true) in either module to access Oracle-specific SQL features
    - Hints, partitioning syntax, analytic function that are not expressible in JPQL

Dimension	Spring Data JDBC	Spring Data JPA
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Comparison

- The N+1 query problem in JPA
  - Loading a list of employees where each employee lazily loads its department
    - Generates one SELECT for the employees plus one SELECT per employee for the department
    - With 100 employees this is 101 queries where 1 join would suffice
    - Always use JOIN FETCH or entity graphs in JPQL, or set relationships to EAGER only with care and measurement

Dimension	Spring Data JDBC	Spring Data JPA
Underlying technology	Spring JdbcTemplate -- no ORM	JPA (Hibernate) -- full ORM
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JPQL

- JPQL
  - Java Persistence Query Language
  - Defined as part of the JPA specification
  - Standard way to express queries against JPA-managed entities.
- Differences
  - SQL operates on tables and columns
    - The physical structures in the database
  - JPQL operates on entity classes and their fields, tje Java objects in the application
    - The query is written in terms of the domain model, not the database schema

```
// SQL -- references the physical table and column names
SELECT e.employee_id, e.full_name, e.salary
FROM   employees e
WHERE  e.department_id = 10
AND    e.status = 'ACTIVE'

// JPQL -- references the Java entity class and field names
SELECT e FROM Employee e
WHERE  e.departmentId = 10
AND    e.status = 'ACTIVE'
```

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Transactions

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Transactions

- The transaction boundary belongs in the service layer
  - Repository methods are not transactional by default in Spring Data JDBC
  - In Spring Data JPA
    - Repository methods run in their own transaction by default, but this is usually too granular
    - A business operation typically involves multiple repository calls that must succeed or fail together
  - The correct pattern is to declare transaction boundaries in the service layer, not the repository layer

```java
@Service
public class EmployeeTransferService {

    private final EmployeeRepository employeeRepository;
    private final AuditRepository auditRepository;

    // @Transactional ensures both repository calls succeed or both roll back
    @Transactional
    public void transferEmployee(Long employeeId, Long newDepartmentId) {
        Employee employee = employeeRepository.findById(employeeId)
                .orElseThrow(() -> new EntityNotFoundException("Employee not found: " + employeeId));

        Long previousDeptId = employee.getDepartmentId();
        employee.setDepartmentId(newDepartmentId);
        employeeRepository.save(employee);

        // Both operations are in the same transaction
        // If this insert fails, the employee update is also rolled back
        auditRepository.save(new AuditRecord(
                employeeId, "TRANSFER", previousDeptId, newDepartmentId
        ));
    }
}
```

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Transactions

- Service layer
  - Methods with @Transactional open a transaction
  - Often calls persistence layer method
- Persistence layer
  - Each persistence method runs in a transaction
  - But we have to integrate the two levels of transactions
  - Done identifying how the persistence layer should or should not integrate with the existing transaction

```java
@Service
public class EmployeeTransferService {

    private final EmployeeRepository employeeRepository;
    private final AuditRepository auditRepository;

    // @Transactional ensures both repository calls succeed or both roll back
    @Transactional
    public void transferEmployee(Long employeeId, Long newDepartmentId) {
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        // Both operations are in the same transaction
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        auditRepository.save(new AuditRecord(
                employeeId, "TRANSFER", previousDeptId, newDepartmentId
        ));
    }
}
```

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Transactions

- Transaction propagation
  - REQUIRED (the default)
    - Join an existing transaction if one exists; create a new one if not
  - REQUIRES_NEW
    - Always create a new transaction, suspending any existing one
    - Used for operations that must commit independently such as audit logging that should survive a rollback of the main transaction
  - SUPPORTS
    - Run in a transaction if one exists; run without one if not
  - NOT_SUPPORTED
    - Always run without a transaction, suspending any existing one

```java
@Service
public class EmployeeTransferService {

    private final EmployeeRepository employeeRepository;
    private final AuditRepository auditRepository;

    // @Transactional ensures both repository calls succeed or both roll back
    @Transactional
    public void transferEmployee(Long employeeId, Long newDepartmentId) {
        Employee employee = employeeRepository.findById(employeeId)
                .orElseThrow(() -> new EntityNotFoundException("Employee not found: " + employeeId));

        Long previousDeptId = employee.getDepartmentId();
        employee.setDepartmentId(newDepartmentId);
        employeeRepository.save(employee);

        // Both operations are in the same transaction
        // If this insert fails, the employee update is also rolled back
        auditRepository.save(new AuditRecord(
                employeeId, "TRANSFER", previousDeptId, newDepartmentId
        ));
    }
}
```

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Transactions

- Example
  - The method from the service layer

```java
// The outer service -- owns the main business transaction
@Service
public class EmployeeTransferService {

    private final EmployeeRepository     employeeRepository;
    private final AuditService           auditService;
    private final NotificationService    notificationService;
    private final SalaryReadService      salaryReadService;

    // REQUIRED (the default)
    // A transaction is started here before the method body runs.
    // Any method called from here that also uses REQUIRED will JOIN this
    // transaction rather than start a new one.
    @Transactional
    public void transferEmployee(Long employeeId, Long newDepartmentId) {

        Employee emp = employeeRepository.findById(employeeId).orElseThrow();
        emp.setDepartmentId(newDepartmentId);
        employeeRepository.save(emp);

        // auditService.recordTransfer uses REQUIRES_NEW -- see below
        // It starts its own independent transaction and commits it immediately
        // If this line succeeds but something below fails and rolls back,
        // the audit record is already committed and will not be rolled back with it
        auditService.recordTransfer(employeeId, newDepartmentId);

        // This line deliberately fails to demonstrate rollback behaviour
        // The employee save above will be rolled back
        // The audit record written above will NOT be rolled back -- it already committed
        if (newDepartmentId.equals(99L)) {
            throw new RuntimeException("Department 99 is not accepting transfers");
        }

        // notificationService.sendEmail uses SUPPORTS -- see below
        // It runs inside the current transaction because one exists
        // but it does not require one -- it would run the same way without one
        notificationService.sendEmail(employeeId, newDepartmentId);
    }
}
```

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Transactions

- Example
  - Method that requires a new transaction

```java
// A service that must commit independently of the caller
@Service
public class AuditService {

    private final AuditRepository auditRepository;

    // REQUIRES_NEW
    // When called from within an existing transaction (like the one in
    // transferEmployee above), Spring suspends that transaction, starts a
    // brand new one for this method, commits it when this method returns,
    // and then resumes the suspended transaction.
    //
    // This means the audit record is committed to the database immediately
    // and permanently -- even if the caller's transaction later rolls back.
    // This is exactly the behaviour needed for audit logging: the fact that
    // a transfer was attempted must be recorded even if the transfer failed.
    @Transactional(propagation = Propagation.REQUIRES_NEW)
    public void recordTransfer(Long employeeId, Long newDepartmentId) {
        auditRepository.save(
                AuditRecord.builder()
                        .employeeId(employeeId)
                        .action("TRANSFER_ATTEMPTED")
                        .newDepartmentId(newDepartmentId)
                        .recordedAt(Instant.now())
                        .build()
        );
        // This transaction commits here, independently of the caller
    }
}
```

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Transactions

- Example
  - Method that requires can work with optional transaction

```java
// A service that works with or without a transaction
@Service
public class NotificationService {

    private final EmailClient emailClient;

    // SUPPORTS
    // If called from within a transaction (as it is here, from transferEmployee),
    // it runs inside that transaction -- though in this case it does not do any
    // database work, so the transaction does not affect its behaviour.
    //
    // If called directly with no active transaction (for example, from a
    // scheduled job or a test), it runs without one rather than creating one.
    //
    // SUPPORTS is appropriate here because sending an email does not need a
    // transaction, but the method should not fail if a caller happens to have
    // one active.
    @Transactional(propagation = Propagation.SUPPORTS)
    public void sendEmail(Long employeeId, Long newDepartmentId) {
        // Email sending does not involve the database at all
        // The propagation setting here is defensive -- it says "I do not need
        // a transaction but I will participate in one if you have started it"
        emailClient.send(
                "Transfer notification for employee " + employeeId,
                "Transferred to department " + newDepartmentId
        );
    }
}
```

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Transactions

- Common mistakes
  - Call repository methods directly from a controller
  - Rely on the repository's own transaction boundary for operations that span multiple repositories
    - Then repository call is its own transaction
    - If the second call fails, the first has already committed and cannot be rolled back
    - The business operation is left in an inconsistent state

```java
// Hint to the database and JPA that no writes will occur
// Allows JPA to skip dirty checking; allows the database to optimize read lock
@Transactional(readOnly = true)
public List<Employee> findActiveEmployees() {
    return employeeRepository.findByStatus("ACTIVE");
}
```

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Transactions

- @Transactional annotation on the service method
  - Ensures that all repository calls within that method share a single transaction
  - Spring wraps the method call, begins a transaction before the method body executes
    - Commits it when the method returns normally
    - Or rolls it back if an unchecked exception propagates out
  - readOnly flag can avoid acquiring row-level locks that a read-write transaction acquires by default

```java
// Hint to the database and JPA that no writes will occur
// Allows JPA to skip dirty checking; allows the database to optimize read lock
@Transactional(readOnly = true)
public List<Employee> findActiveEmployees() {
    return employeeRepository.findByStatus("ACTIVE");
}
```

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Spring Data LDAP

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LDAP

- LDAP
  - Lightweight Directory Access Protocol
  - Protocol for accessing and maintaining distributed directory information services
  - In enterprise environments LDAP directories
    - Store user accounts, group memberships, organizational structures, and authentication credentials
  - Applications use LDAP to
    - Authenticate users by verifying credentials against the directory
    - And to retrieve user attributes like name, email, roles, department

```java
@Entry(
        base = "ou=employees,dc=example,dc=com",  // the subtree to search
        objectClasses = {"top", "person", "organizationalPerson", "inetOrgPerson"}
)
public class LdapEmployee {

    @Id
    private Name distinguishedName;   // the LDAP DN -- the unique identifier in a directory

    @Attribute(name = "cn")
    private String commonName;

    @Attribute(name = "mail")
    private String email;

    @Attribute(name = "departmentNumber")
    private String departmentId;

    @Attribute(name = "memberOf")
    private List<String> groupMemberships;
}
```

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LDAP

- Spring Data LDAP provides
  - The same repository abstraction applied to LDAP directories
  - Object mapping between LDAP entries which have a tree structure of attributes and Java objects
  - Query derivation from method names, exactly as with relational repositories
  - Integration with Spring Security for LDAP-based authentication

```java
public interface LdapEmployeeRepository extends LdapRepository<LdapEmployee> {

    // Find by a specific LDAP attribute
    List<LdapEmployee> findByDepartmentId(String departmentId);

    // Find all members of a specific group
    List<LdapEmployee> findByGroupMembershipsContaining(String groupDn);
}
```

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LDAP

- Typical use in an enterprise application
  - Authentication
    - Verify that a user's credentials are valid against the LDAP directory
  - Authorization
    - Retrieve group memberships to determine what the user is permitted to do
  - User profile
    - load user attributes from the directory rather than storing them in the application's own database

```java
public interface LdapEmployeeRepository extends LdapRepository<LdapEmployee> {

    // Find by a specific LDAP attribute
    List<LdapEmployee> findByDepartmentId(String departmentId);

    // Find all members of a specific group
    List<LdapEmployee> findByGroupMembershipsContaining(String groupDn);
}
```

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Redis

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Redis

- Redis
  - In-memory key-value store that supports a variety of data structures
    - Strings, hashes, lists, sets, sorted sets, and more
  - Extremely fast
    - All data is in memory
    - Operations are single-threaded and avoid locking
    - Typical operation latency is under a millisecond
  - Used in enterprise applications for
    - Caching, session storage, rate limiting, distributed locks, and message queuing (via pub/sub)

```java
@RedisHash("employees")       // key prefix in Redis: "employees:<id>"
@TimeToLive(600)              // entries expire after 600 seconds if not refres
public class CachedEmployee {

    @Id
    private Long employeeId;   // becomes part of the Redis key

    private String fullName;
    private BigDecimal salary;
    private String departmentName;
    private String status;
}

// Repository interface is identical in form to relational repositories
public interface CachedEmployeeRepository
        extends CrudRepository<CachedEmployee, Long> {

    // Spring Data Redis can find by indexed fields
    // (requires @Indexed annotation on the field)
    List<CachedEmployee> findByStatus(String status);
}
```

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Redis

- Redis
  - One of the most important supporting technologies in enterprise Spring Boot applications
  - Most common use case in this context is caching
    - Data that is expensive to retrieve from the database (complex joins, aggregated results, frequently read reference data) is stored in Redis after the first retrieva
    - Subsequent requests are served from Redis without touching the database
  - The @Cacheable and @CacheEvict annotations make this transparent to the application
    - The method body is only executed when the cache does not contain a valid entry

```java
@Service
public class EmployeeService {

    @Cacheable(value = "employees", key = "#id")
    // On first call: fetches from database and stores in Redis
    // On subsequent calls with same id: returns from Redis without hitting dat
    public Employee findById(Long id) {
        return employeeRepository.findById(id)
            .orElseThrow(() -> new EntityNotFoundException("Not found: " + id))
    }

    @CacheEvict(value = "employees", key = "#employee.employeeId")
    // Removes the cached entry when the employee is updated
    public Employee save(Employee employee) {
        return employeeRepository.save(employee);
    }
}
```

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Redis

- Redis
  - Cached data can become stale if the underlying data changes and the cache is not invalidated
  - The TTL provides a backstop
    - Even if explicit eviction is missed, the cached entry will expire and be refreshed from the database within the configured window
    - The appropriate TTL depends on how frequently the data changes and how much staleness is acceptable
    - Reference data that changes quarterly can have a TTL of hours or days
    - User-specific data that changes frequently may need a TTL of seconds or explicit eviction on every write

```java
@Service
public class EmployeeService {

    @Cacheable(value = "employees", key = "#id")
    // On first call: fetches from database and stores in Redis
    // On subsequent calls with same id: returns from Redis without hitting dat
    public Employee findById(Long id) {
        return employeeRepository.findById(id)
            .orElseThrow(() -> new EntityNotFoundException("Not found: " + id))
    }

    @CacheEvict(value = "employees", key = "#employee.employeeId")
    // Removes the cached entry when the employee is updated
    public Employee save(Employee employee) {
        return employeeRepository.save(employee);
    }
}
```

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Rest

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Spring REST

- Exposing data access as RESTful endpoints
- Spring Data REST
  - Inspects the application's Spring Data repositories at startup
  - For each repository, it automatically generates a set of RESTful HTTP endpoints that expose the repository's operations
  - No controller code is required
  - The HTTP layer is generated from the repository interface

```
Given: EmployeeRepository extends JpaRepository<Employee, Long>

Generated endpoints:

GET    /employees              -- find all (with pagination)
GET    /employees/{id}         -- find by ID
POST   /employees              -- create a new employee
PUT    /employees/{id}         -- replace an employee
PATCH  /employees/{id}         -- partially update an employee
DELETE /employees/{id}         -- delete an employee
GET    /employees/search       -- lists available query methods
GET    /employees/search/findByStatus?status=ACTIVE  -- derived query method
```

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Spring REST

- In a normal Spring Boot application
  - A developer writes three things
    - A repository to access the data
    - A service to contain the business logic
    - A controller to expose that logic as HTTP endpoints
  - Spring Data REST eliminates the controller entirely for basic CRUD operations
    - It looks at the repository interfaces in the application at startup
    - For each one it generates the HTTP endpoints automatically
    - No controller class exists in the codebase
    - Spring Data REST creates the HTTP handling layer itself at runtime, using the same proxy and reflection mechanisms that Spring Data uses to generate repository implementations

```
Given: EmployeeRepository extends JpaRepository<Employee, Long>

Generated endpoints:

GET    /employees              -- find all (with pagination)
GET    /employees/{id}         -- find by ID
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GET    /employees/search       -- lists available query methods
GET    /employees/search/findByStatus?status=ACTIVE  -- derived query method
```

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Spring REST

- HATEOAS: the response format
  - Spring Data REST responses follow the HAL (Hypertext Application Language) format
  - Every response includes _links
    - A set of hypermedia links to related resources and available operations
  - This allows clients to navigate the API without hard-coding URLs

```
{
  "employeeId": 1001,
  "fullName": "Jane Smith",
  "salary": 95000.00,
  "_links": {
    "self": { "href": "http://api.example.com/employees/1001" },
    "employee": { "href": "http://api.example.com/employees/1001" },
    "department": { "href": "http://api.example.com/employees/1001/department" }
  }
}
```

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Spring REST

- Annotations allow restricting the generation of some endpoints
- Spring Data REST
  - Is appropriate for internal tooling, rapid prototyping, and simple CRUD APIs where the repository maps cleanly to the resource model
  - Not appropriate for production APIs where response shape, business logic, validation, and security need precise control
    - Use dedicated controllers and service layers instead

```java
// Prevent specific operations from being exposed
@RepositoryRestResource(exported = true)
public interface EmployeeRepository extends JpaRepository<Employee, Long> {

    // Prevent this specific method from being exposed via HTTP
    @RestResource(exported = false)
    void deleteById(Long id);

    // This method IS exposed at /employees/search/findByDepartmentId
    List<Employee> findByDepartmentId(Long departmentId);
}
```

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Questions?